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Comprehensive performance evaluation of steam generating heat pumps for industrial waste heat recovery

Xudong Ma¹, Yuting Wu^{1*}, Yanjun Du¹, Cancan Zhang¹, Biao Lei¹ and Yuanwei Lu¹

Abstract

Steam generating heat pump (SGHP) is a key technology for industrial decarbonization. For comprehensive evaluation of the feasibility and reliability of SGHP in different industrial sector, the work develops a thorough evaluation model for assessing the performance of SGHP by considering waste heat recovery, CO₂ trading value, and pollutant emission cost, in addition to the conventional evaluation criteria. This work presents a thorough comparison of the thermodynamic performance and sustainability of various types of SGHPs across different industrial sector. Additionally, the conflicting relationships between the coefficient of performance (COP) and exergy efficiency are balanced through the application of the technique for order preference by similarity to ideal solution (TOPSIS). The results show that all the indexes of SGHP connected to an open heat pump (SGHPO) with different application scenarios are higher than those of SGHP connected to a flash tank (SGHPF). At the most unfavorable operating condition of the system, the COP minimum value is 1.31 and the exergy efficiency minimum value is 20.42%. These results indicate that replacing the coal-fired boiler with SGHP is feasible and the work could provide theoretical guidance for optimal design and equipment manufacture.

Highlights

- A more comprehensive SGHP valuation model that takes into account the waste heat recovery, CO₂ trading value, and pollutant emission cost is proposed.
- The SGHPs are classified, compared, and analyzed comprehensively, with design and optimization recommendations provided.
- Optimal balance between COP and exergy efficiency of a SGHP using the TOPSIS method.

Keywords Industrial decarbonization, Heat pump evaluation, Steam generating heat pump, Economic analysis, Carbon neutrality

1 Introduction

Many countries globally have set a target of achieving net-zero emissions by 2050 [1, 2]. Especially, the industrial sector contributes the most to total greenhouse gas emissions [3, 4]. Of these, 30% of greenhouse gases are emitted by generating industrial steam [5]. Currently, a significant amount of low-grade waste heat, ranging from 40°C to 100°C, remains underutilized in industrial operations, resulting in recovery rates below 30% [6].

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Steam generating heat pumps (SGHP), as an effective means of energy saving heating, can effectively generate high-temperature steam by recovering medium- and low-temperature waste heat resources and reduce CO₂ emissions [7–9]. Unlike conventional high-temperature heat pumps (HPs), SGHP require incorporation of a steam generator in their design [10]. Figure 1 shows the two main SGHPs: the SGHPO, a closed heat pump connected to an open heat pump, and the SGHPF, a closed

heat pump connected to a flash tank [11]. SGHPF require higher temperatures at the output of the closed heat pump to generate high-temperature steam. Therefore, numerous numerical and experimental studies such as adding subcoolers [12], reheat cycles [13, 14], double flash tank [15], intermediate heat exchangers (IHxs) [16], and injectors [17] have been used to improve the performance of closed HP. Table 1 lists the research of SGHPF.

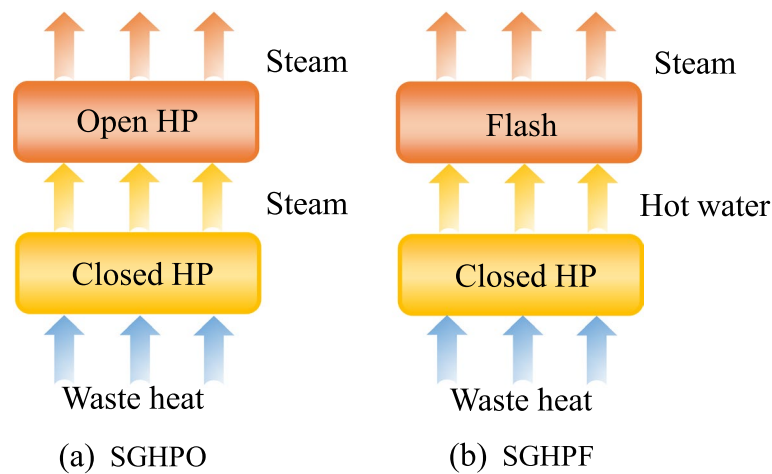


Fig. 1 Classification of SGHPs

Table 1 List of SGHPF research

Methods	References	Type	Refrigerants	Compressor	Heat source temperature /°C	Steam temperature /°C
Simulation	Liu(2015) [18]	Single-stage	R245fa	Twin-screw	60	110
	Liu(2017) [12]	Quasi two- stage	R245fa	Twin-screw	65	-
	Liu(2022) [14]	Single-stage	R1233zd(E)	Twin-screw	30–70	140
	Lu(2022) [10]	Cascade	R1234ze(E)/ R1233zd(Z)	-	30–80	-
	Lu(2019) [19]	Single-stage	R141b	Twin-screw	45–70	100
	Zhang(2022) [15]	Single-stage	R245fa	-	50	-
		Quasi two- stage	R245fa	-	50	-
		Cascade	R404a/R245fa	Twin-screw	5–35	120
	Yan(2021) [20]	Cascade	R404a/R245fa	Twin-screw	5–35	120
	Lee(2017) [21]	Single-stage	R245fa	-	60	104–123
Experiment and simulation	Zhao(2021) [22]	-	-	Piston	50–85	(0.1–0.2MPa)
Experiment	Yang(2018) [17]	Hybrid	R600a	Scroll	-	110 (0.11MPa)
	Martin(2014,2017) [23, 24]	Single-stage	-	-	80–85	184 (1MPa)
						184 (1MPa)
	Kang(2019) [16]	Single-stage	R245fa	Piston	60–70	115–125
	Kobe Steel Corpora- tion(2011,2015) [25, 26]	Two-stage	R245fa	Twin-screw	25–65	100–120(0.1MPa)
	Bellini (2022) [27]	-	R1233zd	Turbocompressor	70	150
	Lee (2021) [28]	Single-stage	R245fa	Piston	50–60	110–120
	Chang(2016) [29]	Single-stage	R245fa	-	-	120(0.1987MPa)

The initial concept of SGHPO, generating steam at temperatures of 135–175°C [30], was first introduced by Kobe Steel in Japan in 2011 [25, 26]. Li et al. [13] proposed a SGHPO that uses two-stage condensation to directly generate low-pressure steam, and then boosted to 0.7MPa by water vapor compression. Yan et al. [20] developed a SGHPO that produces steam above 110°C, achieving an impressive system boiler efficiency of 170%. Zhao et al. [31] found that a SGHPO has a higher COP and a greater construction cost compared to SGHPF. Such systems require vapor recompression enabled by using the water vapor compressor. The potential of sensible and latent heat of water vapor can be maximized. Water vapor compressors must meet the requirements of high compression ratio, volume flow rate and discharge temperature due to the properties of water and system requirements. Therefore, the investment in SGHPO is also relatively high.

For companies, a new equipment is put into operation, not only considering its performance, but also the economic and environmental aspects [30, 32, 33]. Nevertheless, the existing literature lacks a thorough evaluation of the performance, cost-effectiveness, and environmental impact of the two SGHPs. The system complexity of high-temperature heat pumps is expanding, leading to an anticipated increase in the complexity of the SGHP performance evaluation field [34]. One of the significant challenges in this field is the highly combinatorial nature of the various technologies involved. It is imperative to develop a more generalized model encompassing technical parameters for enhancing the accessibility of SGHP [35]. The model should be adaptable to diverse industrial

applications (Waste heat of different temperatures is generated and steam of different temperatures is demanded) featuring different types of SGHPs and contribute to more accurate price calculations and thermodynamic estimations.

Summary, the existing literature has not developed a generalized comprehensive evaluation model to assess the feasibility and reliability of SGHP. However, creating such a model is crucial for effectively evaluating SGHP. Such a model allows fast and accurate prediction of the performance, economics and environmental impact of SGHP in different application scenarios (Waste heat of different temperatures is generated and steam of different temperatures is demanded). Therefore, a generalized evaluation model was developed to comparatively analyze the energy, exergy, economy, and environment (4E) aspects of utilizing SGHP for various application scenarios in this work. The analysis and discussion are conducted by taking into account the full assessment of the waste heat recovery, CO₂ trading value, and pollutant emission cost. The model can be used to generically evaluate SGHPs and establish a theoretical foundation for designing and optimizing them.

2 System description

2.1 System principle

The working principles of the SGHP are illustrated in Fig. 2. The SGHPF primarily comprises an evaporator, condenser, refrigerant compressor, throttling valve, and flash tank. In contrast, the SGHPO includes an evaporator, condenser, refrigerant compressor, water vapor compressor, and throttling valve. Despite the differences

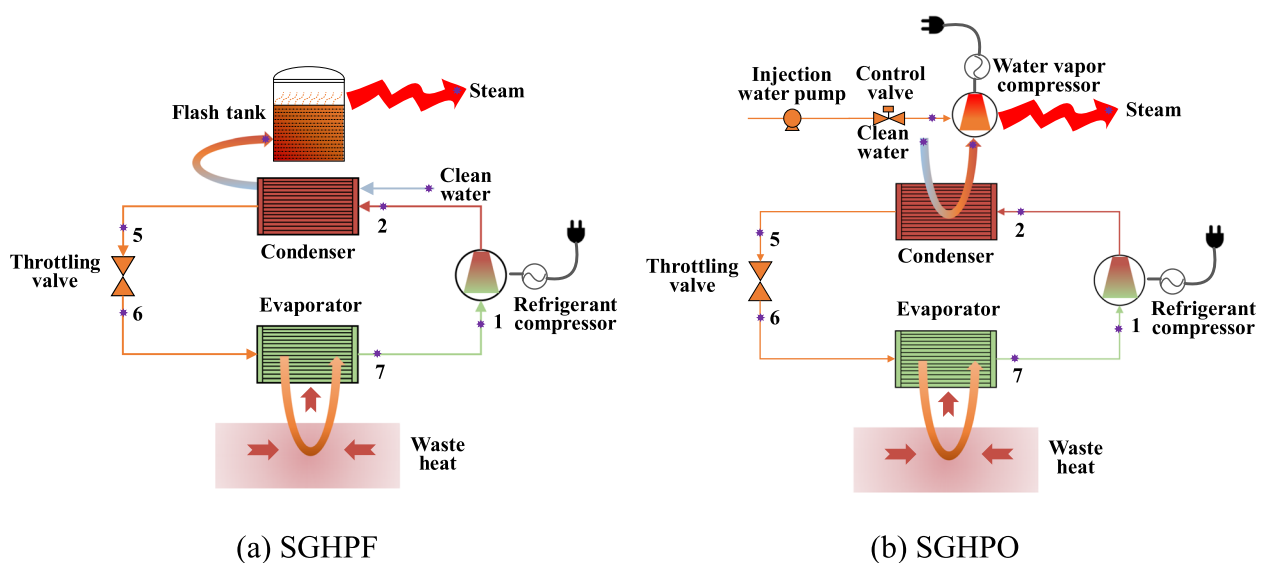


Fig. 2 Schematic diagrams of SGHP

in their components, both systems share a similar closed HP cycle, as depicted in Fig. 3. The evaporator absorbs heat from the waste heat, converting the low-temperature, low-pressure liquid refrigerant into a low-temperature, low-pressure gaseous state. This gaseous refrigerant is then compressed by the refrigerant compressor, which increases its temperature and pressure, transforming it into a high-temperature, high-pressure gaseous refrigerant. Subsequently, in the condenser, the high-temperature, high-pressure gaseous refrigerant releases heat to the liquid water, resulting in a medium-temperature, high-pressure refrigerant. Finally, this medium-temperature, high-pressure refrigerant passes through the throttling valve, where it is converted back into a low-temperature, low-pressure liquid refrigerant, which then re-enters the evaporator to complete the cycle.

The primary difference between the systems lies in the steam generation device. Figure 4 illustrates the steam generation principles of the two SGHPs. SGHPF steam generation path: the water is first pressurized to a pressure corresponding to the output steam, then the water is heated to saturated water at a fixed pressure, after to saturated steam at the corresponding pressure, and finally to

superheated steam through continuous heating. SGHPO steam generation path: liquid water is heated by a closed HP to saturated water, then further heated to saturated steam, and finally pressurized and heated to saturated superheated steam by an open HP. Since the water vapor compressor of the open HP needs to be sprayed with water to ensure that the outlet steam is saturated. Therefore, a multi-stage compression and high efficiency model [36, 37] is used to divide the process into several processes with the same compression ratio. Each process is sprayed with water to absorb the heat of the superheated water vapor and to ensure that the compressed water vapor is saturated at the beginning of each compression process. The exact principle is shown in Fig. 4(a).

2.2 Selection of refrigerant

Refrigerant is a key factor in HP performance. For SGHPs, following principles should be considered for choosing the refrigerant: (1) low GWP. (2) zero/low ODP (3) excellent thermodynamic properties and practicality. (4) non-toxic and non-flammable. (5) low cost. (6) short atmospheric lifetime. (7) high critical temperature. Based on these principles, the properties of the

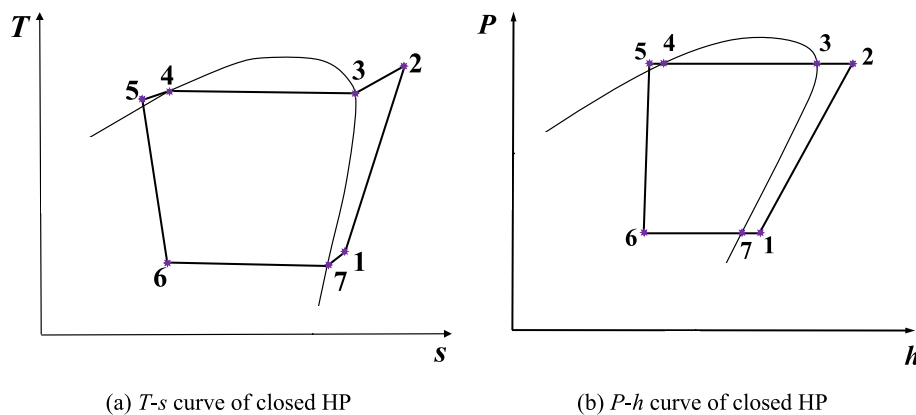


Fig. 3 Closed HP principle

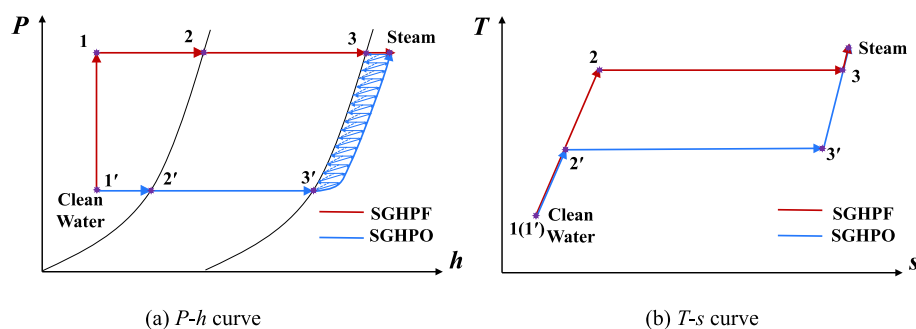


Fig. 4 Steam-generation path

refrigerants selected are shown in Table 2. The selection included one HFC refrigerant and four HFO refrigerants. Under the Kigali Amendment to the Montreal Protocol, a phased reduction in the use of HFCs is mandated, beginning in 2024 and reaching 20% of the baseline by 2034 [38]. This regulation permits the limited continued use of HFC refrigerants in critical sectors. Additionally, HFO refrigerants, which are being developed as potential alternatives to R245fa, are currently expensive. Moreover, according to ECHA latest proposal for PFAS regulation, most current lower GWP HFC/HFO alternative refrigerants would face a ban [39]. Single-component refrigerants, such as R134a, R143a, R1234yf, R1234ze(E), R1336mzz(E), R1336mzz(Z), R1224yd, and R1233zd(E), fall under the definition of PFAS in refrigerants [39]. In this study, the charge of R245fa refrigerant was low, resulting in a minimal environmental impact. Consequently, the negligible environmental impact of R245fa should be considered in the context of overall contribution of the system to environmental sustainability. Given these factors, it is crucial to carefully assess the economic and technical feasibility of transitioning to HFO refrigerant before its introduction. Moreover, the literature review revealed that a significant number of high temperature heat pumps continue to utilize R245fa. Therefore, to reach a universal conclusion, this work employs R245fa refrigerant as the closed HP refrigerant.

2.3 Model validation

The results of closed HP simulated values in this paper, as illustrated in Fig. 4, are compared with the values from the literature by ZHANG, et al. (2022) [15]. It is essential to note that the simulated values in this study are consistently smaller than the literature values. The maximum error observed is 16.23%, while the minimum error is 5.63%. Notably, the error rate demonstrates a decreasing trend as the condensation temperature increases. Further details regarding the model utilized in this research can be found in Sect. 3. The main reason for the error is that the literature values use theoretically compressor data, while the simulation in this paper use measured single-screw compressor data, and the measured compressor

will produce many engineering errors. Table 3. presents the primary parameters of the self-developed single screw compressor. This also leads to a lower efficiency of the measured compressor. Therefore, the model in this paper has accuracy (Fig. 5).

3 Methods

The following assumptions were made for the sake of simplifying the modeling process and managing computational complexity [40]. First and foremost, the system is assumed to be in a steady state. Secondly, the modeling process excludes considerations of piping heat loss and pressure drop losses. Additionally, the throttling process is simplified to an isenthalpic process. Finally, the reference temperature is set at room temperature, where $T_0 = 20^\circ\text{C}$ and $P_0 = 0.101325\text{MPa}$ as the state of reference. The modeling process was achieved using MATLAB software and the specific calculations are shown in Fig. 6. The refrigerant properties are calculated by a subroutine of NIST REFPROP 10.0 [41].

3.1 Thermodynamic performance analysis

During steady-state operation, the mass of steam produced from clean water remains constant.

Table 3 Single-screw compressor main technical parameters

Parameters		value
Screw diameter		>147mm
Star wheel diameter		147mm
Rotation speed		2880r/min
Volumetric flow rate		167.88m ³ /h
Number of screw heads		6
Number of star wheel teeth		11
Volumetric	Length	1310mm
	Width	460mm
	Height	460mm
Weight		430kg
Motor voltage		380V

Table 2 Selected key physical properties of refrigerants

Category	Refrigerant	Critical temperature/°C	Critical pressure/bar	ODP	GWP	SG	NBP/°C	MW/g·mol ⁻¹
HFOs	R1233zd(E)	166.5	36.2	0.00034	1	A1	18	130.5
	R1224yd(Z)	155.5	33.3	0.00012	< 1	A1	14	148.5
	R1234ze(Z)	150.1	35.3	0	1	A2L	9.8	114.0
	R1336mzz(Z)	171.3	29.0	0	2	A1	33.4	164.1
HFCs	R245fa	154.01	36.51	0	1030	B1	15	134.05

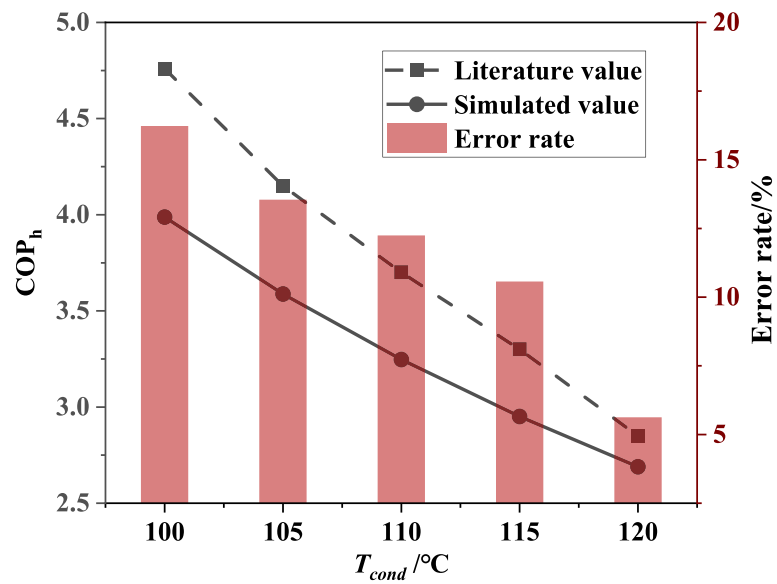


Fig. 5 Comparing closed HP simulated values with values from the literature

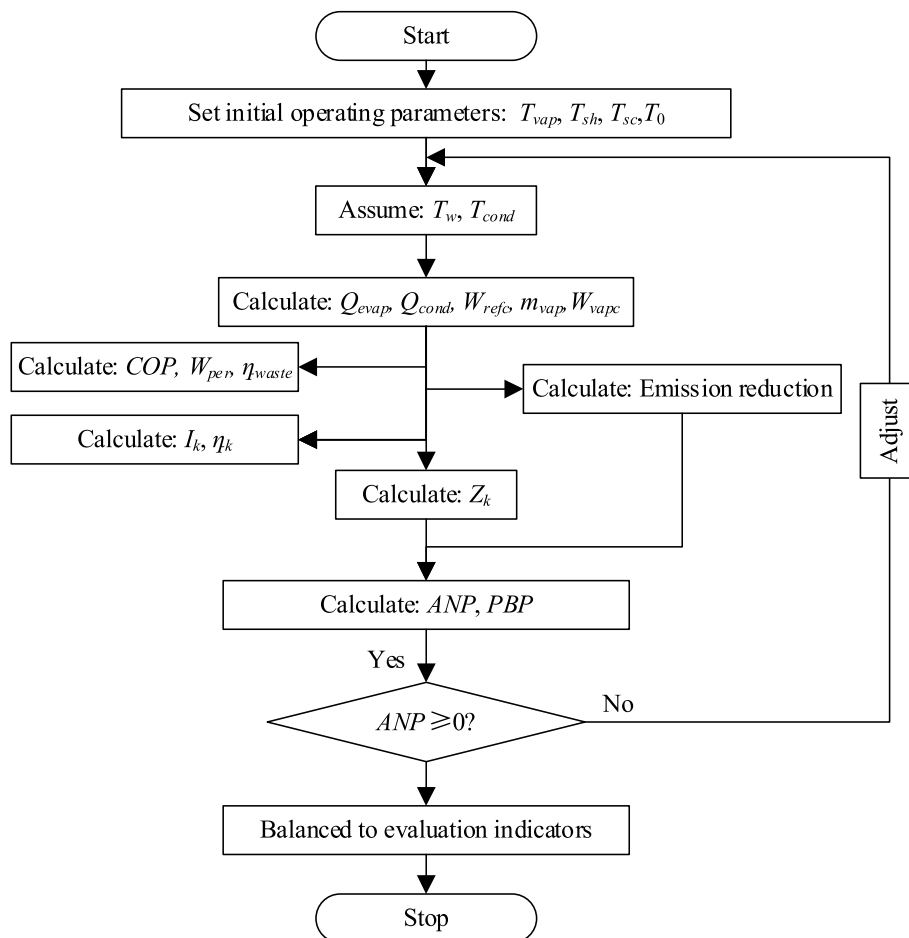


Fig. 6 Flow chart of the model calculation programme

$$M_l = M_{vap} \quad (1)$$

Conservation of system energy:

$$m_l h_l + W_{refc} + W_{vpc} + Q_{e\ vap} = m_{vap} h_{vap} \quad (2)$$

The exergy destruction and efficiency of SGHP in steady state can be defined by the equation [42]:

$$I_k = \Sigma Ex_{in,k} - \Sigma Ex_{out,k} \quad (3)$$

$$\eta_k = \frac{Ex_{revenue}}{Ex_{payment}} \quad (4)$$

Evaporator:

$$Q_{evap} = C_{p,w} m_w (T_{w,in} - T_{w,out}) \quad (5)$$

$$I_{evap} = m_f [(h_{evap,in} - h_{evap,out}) - T_0 (s_{evap,in} - s_{evap,out})] + Q_{evap} \left[1 - \frac{T_0}{(T_{w,in} - T_{w,out}) / \ln (T_{w,in} / T_{w,out})} \right] \quad (6)$$

$$\eta_{evap} = \frac{m_f [(h_{evap,out} - h_{evap,in}) - T_0 (s_{evap,out} - s_{evap,in})]}{Q_{evap} \left[1 - \frac{T_0}{(T_{w,in} - T_{w,out}) / \ln (T_{w,in} / T_{w,out})} \right]} \quad (7)$$

Refrigerant compressor:

$$W_{refc} = m_f (h_{cond,in} - h_{evap,out}) \quad (8)$$

$$I_{refc} = W_{refc} + m_f [(h_{evap,out} - h_{cond,in}) - T_0 (s_{evap,out} - s_{cond,in})] \quad (9)$$

$$\eta_{refc} = \frac{m_f [(h_{evap,out} - h_{cond,in}) - T_0 (s_{evap,out} - s_{cond,in})]}{W_{refc}} \quad (10)$$

Condenser:

$$Q_{cond} = m_f (h_{cond,in} - h_{cond,out}) = m_l (h_{l,out} - h_{l,in}) \quad (11)$$

$$I_{cond} = m_f [(h_{cond,in} - h_{cond,out}) - T_0 (s_{cond,in} - s_{cond,out})] - Q_{cond} \left[1 - \frac{T_0}{(T_{l,out} - T_{l,in}) / \ln (T_{l,out} / T_{l,in})} \right] \quad (12)$$

$$\eta_{cond} = \frac{Q_{cond} \left[1 - \frac{T_0}{(T_{l,out} - T_{l,in}) / \ln (T_{l,out} / T_{l,in})} \right]}{m_f [(h_{cond,in} - h_{cond,out}) - T_0 (s_{cond,in} - s_{cond,out})]} \quad (13)$$

Throttling valve:

$$m_f h_{cond,out} = m_f h_{evap,in} \quad (14)$$

$$I_{tv} = m_f T_0 (s_{evap,in} - s_{cond,out}) \quad (15)$$

$$\eta_{tv} = \frac{(h_{evap,in} - h_0) - T_0 (s_{evap,in} - s_0)}{(h_{cond,out} - h_0) - T_0 (s_{cond,out} - s_0)} \quad (16)$$

Water vapor compressor

$$W_{vpc} = m_{sup,vap} h_{sup,vap} - m_{sat,vap} h_{sat,vap} \quad (17)$$

$$I_{vpc} = W_{vpc} + m_{vap} [(h_{sat,vap} - h_{sup,vap}) - T_0 (s_{sat,vap} - s_{sup,vap})] \quad (18)$$

$$\eta_{vpc} = \frac{m_{vap} [(h_{sup,vap} - h_{sat,vap}) - T_0 (s_{sup,vap} - s_{sat,vap})]}{W_{vpc}} \quad (19)$$

SGHP coefficient of performance (COP):

$$COP = \frac{m_{vap} h_{vap} - m_l h_l}{W_{refc} + W_{vpc}} \quad (20)$$

Power consumption per unit of steam:

$$W_{per} = (W_{refc} + W_{vpc}) / M_{vap} \quad (21)$$

Waste heat utilization rate:

$$\eta_{waste} = \frac{Q_{evap}}{m_{vap} h_{vap} - m_l h_l} \quad (22)$$

Total exergy efficiency:

$$\eta_{tot} = \frac{Ex_{revenue}}{Ex_{payment}} = \frac{W_{vpc} + W_{refc} - I_{tot}}{W_{vpc} + W_{refc}} = 1 - \frac{I_{tot}}{W_{vpc} + W_{refc}} \quad (23)$$

3.2 Sustainability analysis

Various metrics, including PRR [43] and EPC [44], can evaluate the economics of heat pumps. This study, how-

ever, emphasizes the feasibility and reliability of using SGHP to replace coal-fired boilers in enterprises. Therefore, the economics of SGHP were analyzed using the equivalent annual cost (EAC) [7], which considers both the annual capital investment (Z_{CI}) and the annual operation and maintenance costs (Z_{OM}). The initial investment cost rate for each component can be expressed as:

$$Z_k^{CI} = CRF \cdot Z_k^{CI} \quad (24)$$

In this work, the interest rate (i) is taken as 15%, and the life cycle of the system (n) is taken as 10 [40, 45]. The

capital recovery factor (CRF) for the equipment depends on both the interest rate and the life cycle of the system. It represents the ratio of the annual recovery amount to the initial investment spread evenly over the life cycle according to the compound interest rate.

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (25)$$

The largest investment in the SGHP is the purchase of the water vapor compressor (Z_{vapc}), refrigerant compressor (Z_{refc}), condenser (Z_{cond}), and evaporator (Z_{evap}). The cost of extra equipment (Z_{other}) and the throttling valve constitutes 15% of the main equipment cost [46].

The total investment cost is calculated as follows:

$$Z_{CI} = Z_{refc} + Z_{vapc} + Z_{cond} + Z_{evap} + Z_{other} \quad (26)$$

The investment cost for a refrigerant compressor and water vapor compressor can be calculated using the expression [47]:

$$Z_{refc} = 7364 \cdot m_f \cdot \left(\frac{P_2}{P_1}\right) \cdot \left(\frac{\psi_{refc}}{1 - \psi_{refc}}\right)^{0.7} \quad (27)$$

Here, ψ_{refc} represents the isentropic efficiency of the refrigerant compressor, and P_1 and P_2 represent the suction and discharge pressures of compressor, respectively.

The investment cost of heat exchangers is as follows [48–50]:

$$Z_{hx} = Z_{R,hx} \left(\frac{A_{hx}}{A_R}\right)^{0.6} \quad (28)$$

The subscript R represents the reference value.

The heat exchanger area can be determined with the formula:

$$A_{hx} = \frac{Q_{hx}}{KF\Delta T_{LMTD}} \quad (29)$$

In this context, K stands for the total heat transfer coefficient and F represents the correction factor for the logarithmic mean temperature difference in heat exchangers. Table 4 lists the parameters.

utilized in heat exchanger calculations.

The logarithmic mean temperature difference is calculated using the following equation:

$$\Delta T_{LMTD} = \frac{\Delta T_{\max} - \Delta T_{\min}}{\ln(\Delta T_{\max}/\Delta T_{\min})} \quad (30)$$

The formula below provides the equivalent annual cost of each component:

$$Z_k = Z_k^{CI} + Z_k^{OM} \quad (31)$$

Table 4 Specific parameters of the selected heat exchangers

Parameters	Unit	Value	References
K	W/(m ² ·k)	1200	[51]
F	/	0.93	[52]
$Z_{R,k}$	\$	16,000	[48–50]
A_k	m ²	100	[48–50]
φ	/	0.06	[49, 53]

Table 5 Economic benefit calculation parameters

Factor	Unit	Value	References
Steam price	\$/t	27.42	[54]
Electricity price	\$/kW·h	0.05	[45]
Waste heat recovery price	\$/kW·h	0.05	[55]

Annual operating cost per unit:

$$Z_k^{OM} = \varphi \cdot Z_k^{CI} \quad (32)$$

The use of SGHP instead of boilers to generate steam reduces CO₂, SO₂ and NO_x emissions and save penalty costs for emissions. The annual net profit (ANP) can be calculated as follows [40]:

$$ANP = m_{vap} \cdot N \cdot Z_{vap} + Q_{evap} \cdot N \cdot Z_{waste} + Z_{CO_2} + Z_{SO_2} + Z_{NO_2} - W \cdot N \cdot Z_{electricity} \quad (33)$$

where N represents the annual operating time of the SGHP, usually the operating time of the plant in a company is 8000 h/year. m_{vap} represents the mass flow rate of steam, W represents the input power of the SGHP, Z_{vap} and $Z_{electricity}$ represent the unit price of the corresponding steam and electricity, respectively. Z_{waste} represents the unit price of waste heat recovery.

The payback period (PBP) can be calculated as follows:

$$PBP = \frac{Z_{tot}}{ANP} \quad (34)$$

Table 5 outlines the parameters for calculating the economic benefits.

This section primarily focuses on the emissions of CO₂, SO₂ and NO_x [56]. It can be expressed as follows:

$$m_{gas} = a_{gas} \cdot m_{coal} - b_{gas} \cdot W \quad (35)$$

where a represents the emission conversion factor for standard coal and b represents the emission conversion factor for standard electricity. m_{coal} represents the quality of standard coal and can be calculated as follows:

$$m_{coal} = Q_{vap} \cdot N / Q_{standard} \quad (36)$$

Table 6 Calculate parameters of the emission gas

Factor	Unit	CO ₂	SO ₂	NO _x	References
a	kg/kg	2.493	0.075	0.0375	[57]
b	kg/kW·h	0.997	0.030	0.015	[57]
c	\$/t	7.29	344.75	689.50	[58, 59]

where $Q_{standard}$ represents the standard coal heat (29,307.6kJ/kg), and Q_{vap} represents the heat of steam generated.

The environmental benefits can be calculated as follows:

$$Z_{gas} = m_{gas} \cdot c_{gas} \quad (37)$$

The c represents the price per unit of penalty cost, with corresponding values listed in Table 6.

3.3 Balanced approach to evaluation indicators

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), is a prevalent comprehensive evaluation technique that aims to approximate the ideal solution ordering. By analyzing the original data, this method generates results that accurately depict the variations among different evaluation programs [60]. The distance from the i^{th} solution on the η_{tot} curve to the ideal and non-ideal solutions is specifically defined as [61]:

$$d_{i+} = \sqrt{(COP_i^n - COP^{n,ideal})^2 + (\eta_{tot,i}^n - \eta_{tot}^{n,ideal})^2} \quad (38)$$

$$d_{i-} = \sqrt{(COP_i^n - COP^{n,non-ideal})^2 + (\eta_{tot,i}^n - \eta_{tot}^{n,non-ideal})^2} \quad (39)$$

COP_i^n and $\eta_{tot,i}^n$ are the dimensionless values of the COP and η_{tot} .

$$COP_i^n = \frac{COP_i - COP_i^{\min}}{COP_i^{\max} - COP_i^{\min}} \quad (40)$$

$$\eta_{tot,i}^n = \frac{\eta_{tot,i}^{\max} - \eta_{tot,i}}{\eta_i^{\max} - \eta_i^{\min}} \quad (41)$$

The proximity index of the i^{th} solution is defined as.

$$Y_i = \frac{d_{i-}}{d_{i-} + d_{i+}} \quad (42)$$

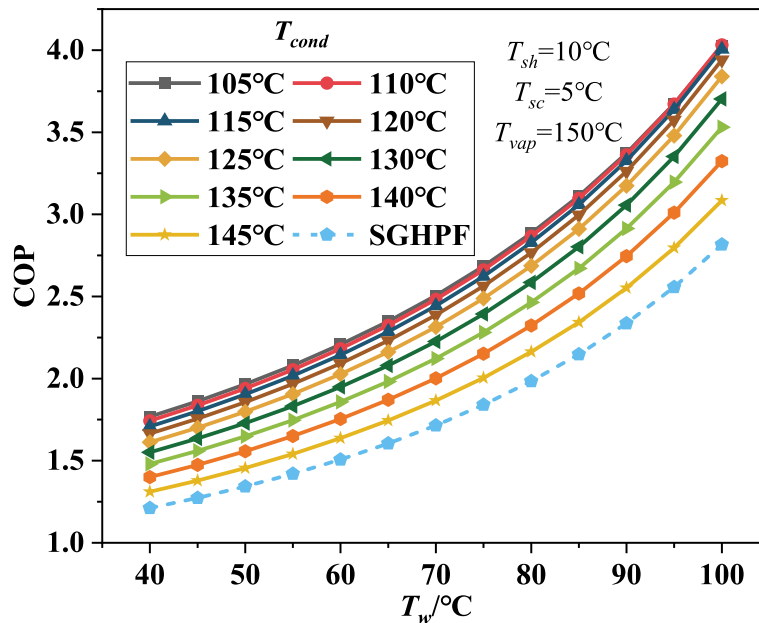
The final optimal solution is determined by finding the solution with the highest proximity index Y .

4 Results and discussion

4.1 Energy performance comparison

4.1.1 Effect of waste heat temperature

For evaluating the performance of the two SGHPs, a comparative analysis of the COP as the waste heat temperature varies was performed, as shown in Fig. 7. The results showed that the COP of both SGHPO and SGHPF increased as the heat source temperature increased. The COP of SGHPO is always higher than SGHPF, and the COP improvement effect is lower as the condensation temperature of SGHPO increases, with a minimum improvement rate of 8.17%~9.55% ($T_{cond}=145^\circ\text{C}$). As


Fig. 7 Comparison of COP of SGHPF and SGHPO with different condensation temperature

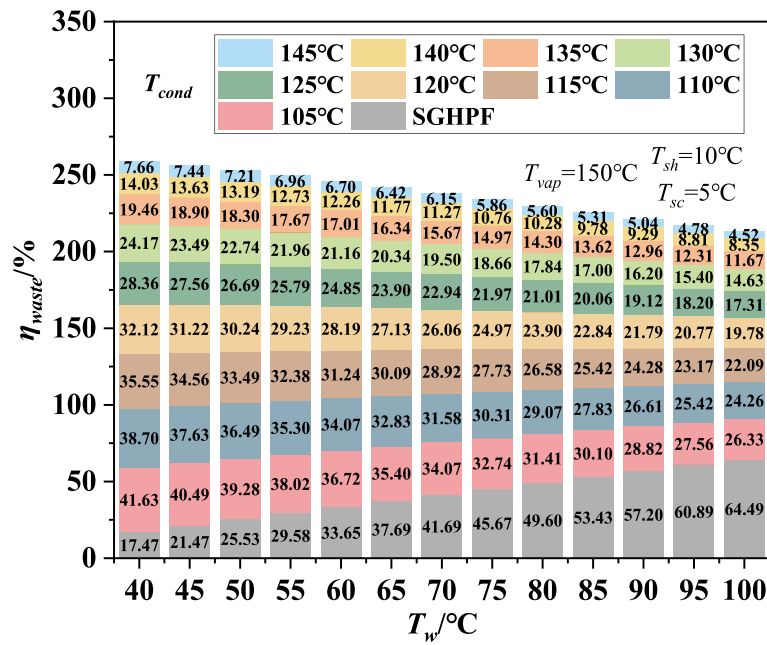


Fig. 8 Comparison of the increase in waste heat utilization rate of SGHPO with different condensation temperatures compared to SGHPF

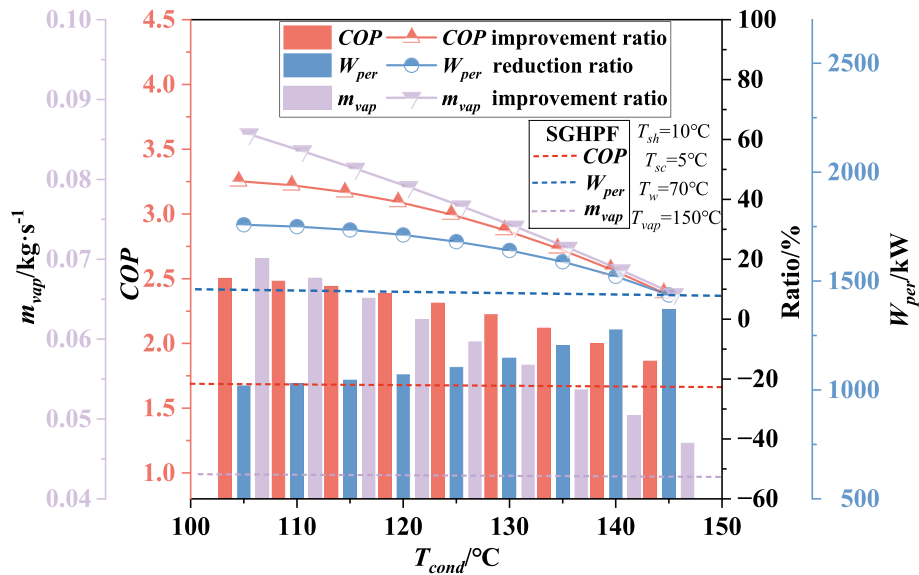
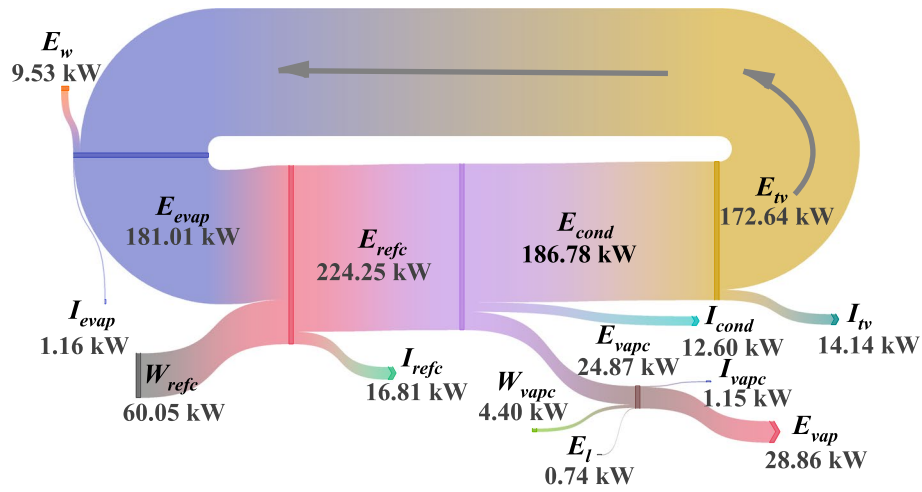
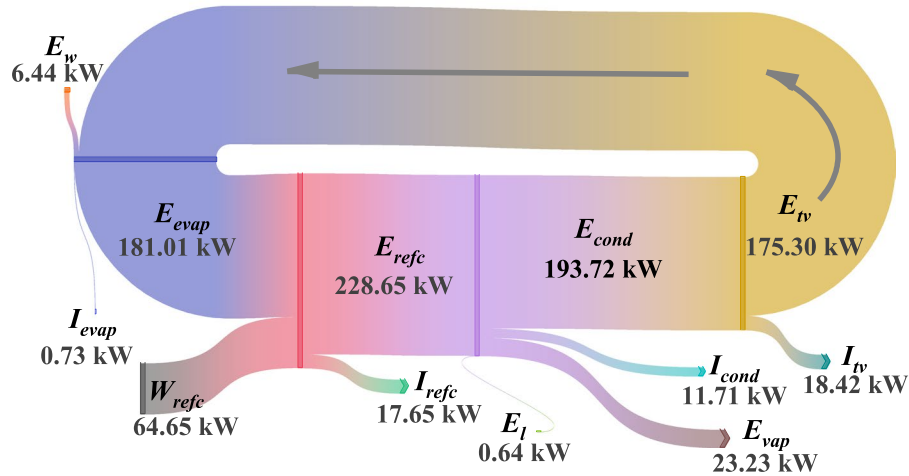


Fig. 9 Steam mass flow rate, COP and power consumption per unit of steam for SGHPO with different condensation temperatures vs. SGHPF

the heat source temperature rises, both waste heat utilization rates of SGHP increase, leading to an increase in the COP. Figure 8 shows the correlation between waste heat utilization rate and waste heat temperature. However, as the condensation temperature of SGHPO increases, the effect of improving the waste heat utilization rate of SGHPO compared to SGHPF decreases. Therefore, as the condensation temperature of SGHPO increases, the COP decreases.

4.1.2 Effect of condensation temperature

When the waste heat temperature remains constant, Fig. 9 shows the steam mass flow rate, COP and power consumption per unit of steam for SGHPO at varying condensation temperatures compared to SGHPF. An observation from the data is that with the increase in condensation temperature, there is a decrease in the rates of COP improvement, steam mass flow rate improvement, and reduction rate of power consumption per unit


(a) SGHPO ($T_{cond}=140^{\circ}\text{C}$)


(b) SGHPF

Fig. 10 Exergy flow of the SGHP ($T_w=70^{\circ}\text{C}$)

of steam for SGHPO in comparison to SGHPF. This is because, when the waste heat temperature stays constant, raising the condensation temperature results in higher power consumption for the refrigerant compressor and a lower waste heat utilization ratio, leading to a decrease in the heat capacity of the SGHP. Therefore, the steam mass flow rate decreases. Since the steam temperature remains constant, the water vapor compressor power consumption decreases, but the decrease is small. As a result, the COP decreases and the power consumption per unit of steam increases. When the condensation temperature is 105°C , the COP of SGHPO is 2.51, while the COP of SGHPF is only 1.71.

Summary, the thermodynamic performance of SGHPO surpasses that of SGHPF. The COP, waste heat utilization ratio, and steam mass flow rate of SGHPO exhibit positive correlations with the waste heat temperature and negative correlations with the condensation temperature. Conversely, the trend in power consumption per unit of steam shows an opposite relationship to these factors.

4.2 Exergy performance comparison

An exergy analysis is conducted to show the revenue and payment exergy in the SGHP. Figure 10 shows the exergy

flow of the SGHP under a fixed waste heat temperature of 70°C. Although the total compressor input power of the SGHPO and the refrigerant compressor input power of the SGHPF are basically equal, the waste heat input exergy and output exergy (E_{vap}) of the SGHPO are larger than that of the SGHPF. Therefore, the SGHPO consistently demonstrates higher exergy efficiency and heat exergy output than the SGHPF.

4.2.1 Effect of waste heat temperature

The comparison of the total exergy efficiency of SGHPO and SGHPF at different waste heat temperatures is shown in Fig. 11. An observed pattern is that the total exergy efficiency of all systems generally exhibits an upward trend followed by a decrease as the waste heat temperature rises. Notably, the peak values predominantly occur around the waste heat temperatures of 60°C or 65°C. And the total exergy efficiency of SGHPO with different condensation temperatures is higher than that of SGHPF. The total exergy efficiency of SGHPF is 19.40%~25.04%, while SGHPO can reach 42.59%~47.72% at the condensation temperature of 105°C.

The percentage of throttling valve and refrigerant compressor exergy destruction decreases, and the exergy efficiency both increase slightly when the waste heat temperature increases, as illustrated in Fig. 12. Consequently, this results in a decrease in the total exergy efficiency. Conversely, the condenser, water vapor compressor, and evaporator show an increased percentage of exergy destruction, with the water vapor compressor

exergy efficiency remaining relatively constant while the evaporator and condenser exergy efficiency increase. For instance, the evaporator exergy efficiency demonstrates a significant increase from 65.76% to 92.95% within a waste heat temperature range from 40°C to 100°C, contributing to an overall enhancement in the total exergy efficiency. As a result of these trends, the total exergy efficiency initially rises and subsequently declines due to the combined effects of these components.

4.2.2 Effect of condensation temperature

When the waste heat temperature is consistent, the total exergy efficiency and total exergy destruction of SGHPO compared to SGHPF vary with different condensation temperatures, as illustrated in Fig. 13. It is noted that the improvement rate in total exergy efficiency and reduction rate in total exergy destruction diminish as the condensation temperature rises. For instance, at a condensation temperature of 105°C, the total exergy efficiency of SGHPO reaches 47.63%, significantly higher than the 24.97% efficiency of SGHPF. This disparity is attributable to the rise in refrigerant compressor power consumption and the decline in waste heat utilization ratio with increasing condensation temperature, thereby causing a decline in the overall exergy efficiency of SGHPO.

Summary, both SGHPO and SGHPF total exergy efficiency increase and then decrease with waste heat temperature, with a peak value. The SGHPO total exergy efficiency decreases with condensation temperature, but is always greater than SGHPF.

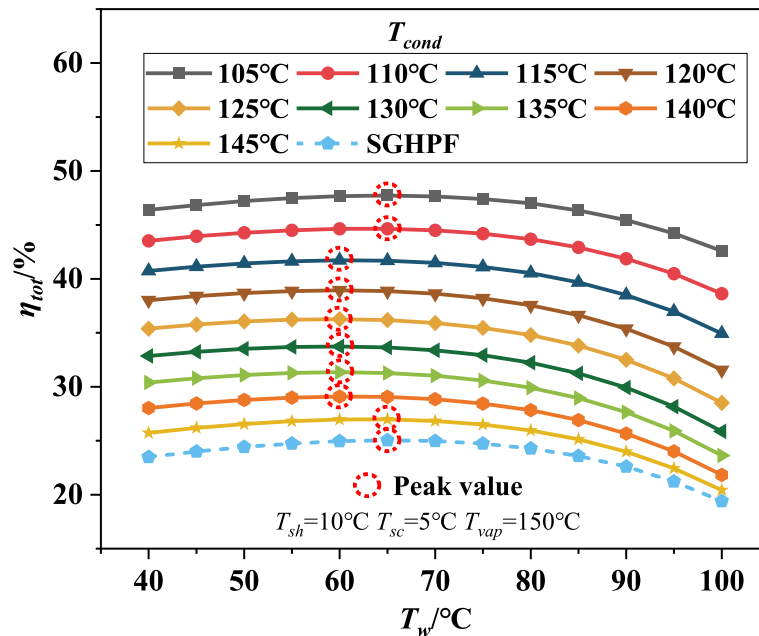


Fig. 11 Comparison of total exergy efficiency of SGHPF and SGHPO with different condensation temperature

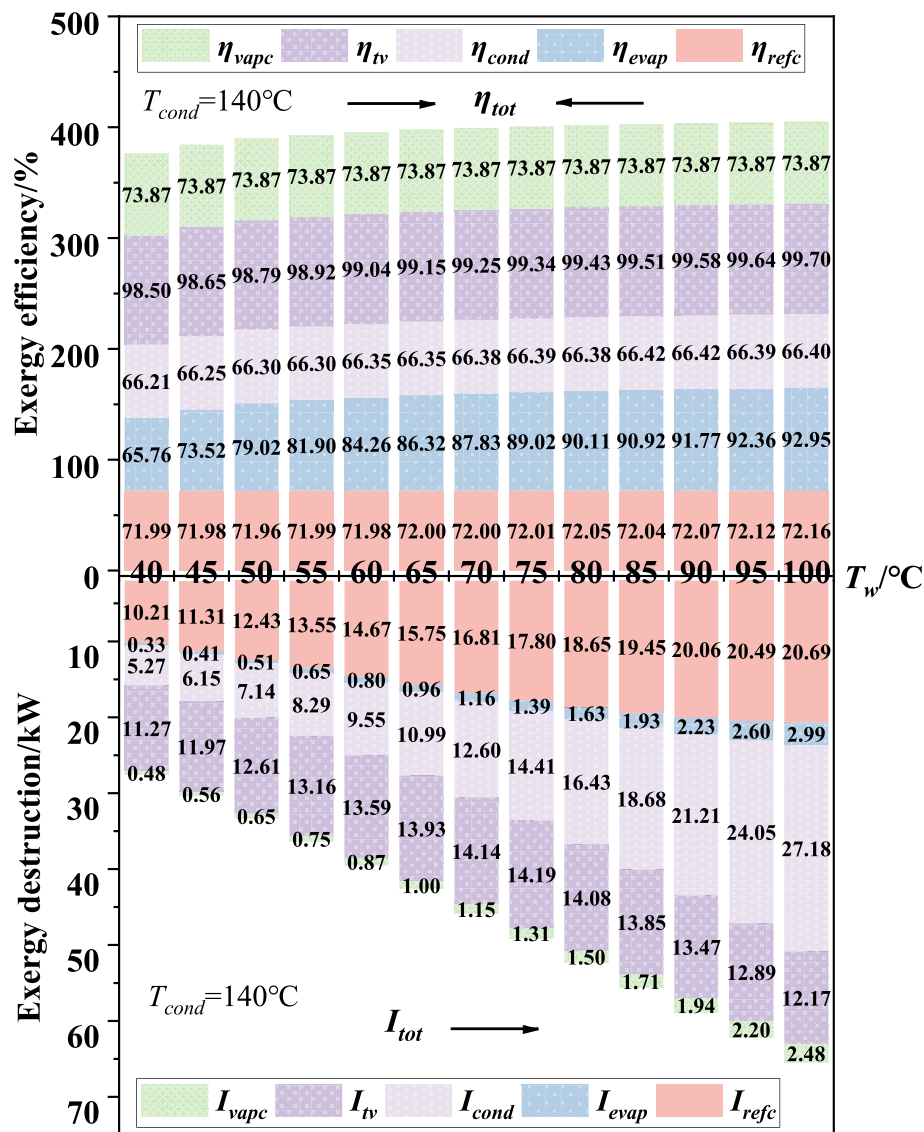


Fig. 12 Effect of heat source temperature on the exergy evaluation of the SGHPO

4.3 Economic analyses comparison

Figure 14 shows the comparison of ANP and PBP of SGHPO and SGHPF with waste heat temperatures. It can be seen that the ANP of SGHPO with different condensation temperatures is higher than that of SGHPF when generating steam at the same waste heat temperature. Moreover, the ANP gradually increases with the increased waste heat temperature and the decreased condensation temperature. The SGHPO generates ANP only when the condensation temperature is below 125 °C at a waste heat temperature of 40 °C. The maximum and minimum PBP values are 1.33 years (at 125 °C) and 0.14 years (at 105 °C), respectively. The minimum waste heat temperature for SGHPF to generate ANP is 70 °C,

at which case the PBP is 1.26 years. The maximum PBP for the SGHPO in the case is only 0.18 years (condensation temperature is 145 °C). As the waste heat temperature rises, various SGHPs have a consistent PBP of 0.02 years ANP with minimal variation. Therefore, when the waste heat temperature is high, both SGHPs can produce significant economic gains. And the difference in PBP is very small, but SGHPO can generate higher ANP. And when the waste heat temperature is lower, both ANP and PBP of SGHPO are much higher than those of SGHPF. Therefore, the lower the condensation temperature and the higher waste heat temperature, the better the system benefits.

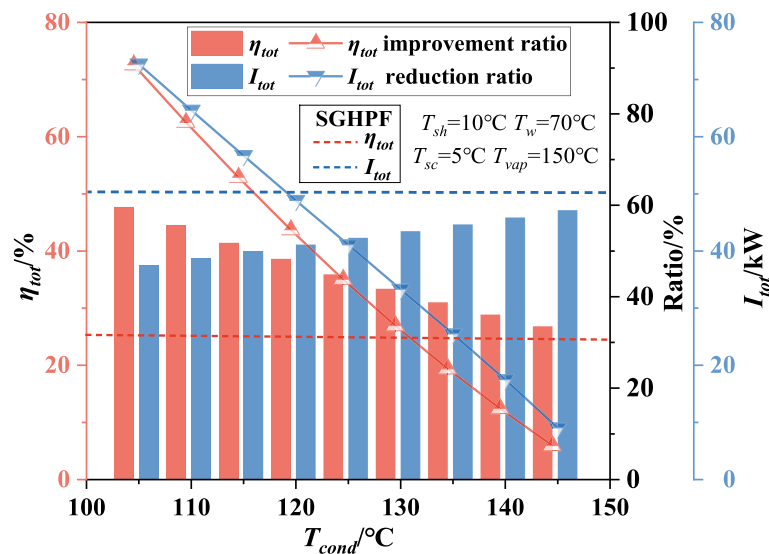


Fig. 13 Comparison of total exergy evaluation of SGHPF and SGHPO with different condensation temperature

4.4 Environmental analyses comparison

Figure 15 illustrates the comparison of annual emission reduction between SGHPO and SGHPF at varying condensation temperatures. The annual reduction of pollutant gases increased as the waste heat temperature rose for both SGHPO and SGHPF. The annual emission reductions of pollutant gases of SGHPO with different condensation temperatures were higher than those of SGHPF, and the annual emission reductions of SGHPO decreased gradually with the condensation temperature increasing. The annual emission reduction of SGHPF is the minimum value of SGHP, which is 127.09 t. The SGHP can achieve a maximum annual emission reduction of 1059.88t at a waste heat temperature of 100 °C, with the condensation temperature of SGHPO set at 105 °C. Among the three pollutants, CO₂ has the highest share, with an average share of 96%.

Summary, both SGHPs can decrease pollutant emissions. This reduction is directly linked to the waste heat temperature and inversely related to the condensation temperature of the SGHPO.

4.5 Balanced to evaluation indicators

Since COP is positively proportional to ANP, this paper mainly seeks the optimal balance between energy and exergy. Figure 16 shows the evaluation indicator balance results of SGHPO at condensation temperature of 140 °C. It can be seen that as the COP increases from 1.40 to 3.32, the exergy efficiency first increases from 28.03% to 29.10% and then decreases to 21.83%. The TOPSIS method first located the coordinates of the ideal and non-ideal points. Then, the distance to the ideal point (d_+),

the distance to the non-ideal point (d_-), and the neighborhood index Y were calculated for each non-dominated solution on the exergy efficiency curve. Figure 16 illustrates that the proximity index fluctuates with the COP, first increasing and then decreasing. In this case, the final evaluation indicator balance solution with maximum proximity index Y of 0.877 is localized when COP and exergy efficiency are 2.15 and 28.45%, respectively.

Based on the above calculations, balanced to evaluation indicators were developed for SGHPO and SGHPF with different condensation temperatures, and the specific results are shown in Fig. 17. Both COP and exergy efficiency of the evaluation indicator balance solution of SGHPO are larger than SGHPF. For SGHP that recovers waste heat in a wide temperature range, the optimal balance of performance of SGHPO are better than SGHPF. The optimal COP and exergy efficiency of SGHPO at a condensation temperature of 105 °C are 2.89 and 47.00%, respectively.

5 Conclusion

The work develops a generalized model for evaluating SGHPs, taking into account the value of waste heat recovery, CO₂ trading value, and pollutant emission cost. The model is used to assess the feasibility and reliability of using SGHP in various scenarios. The TOPSIS method was employed to optimize the design and operation of the SGHP by balancing COP and exergy efficiency. Recommendations for system optimization under varying working conditions are presented in Fig. 18. The main conclusions can be summarized as follows:

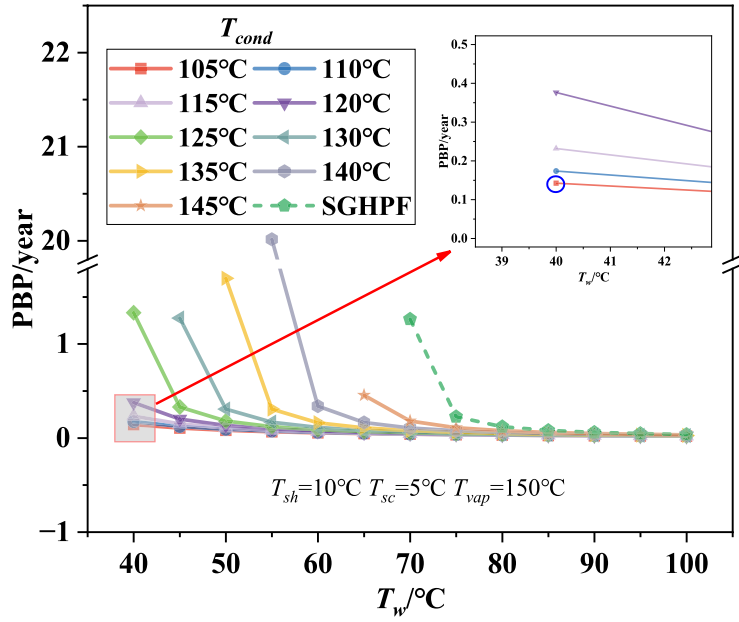
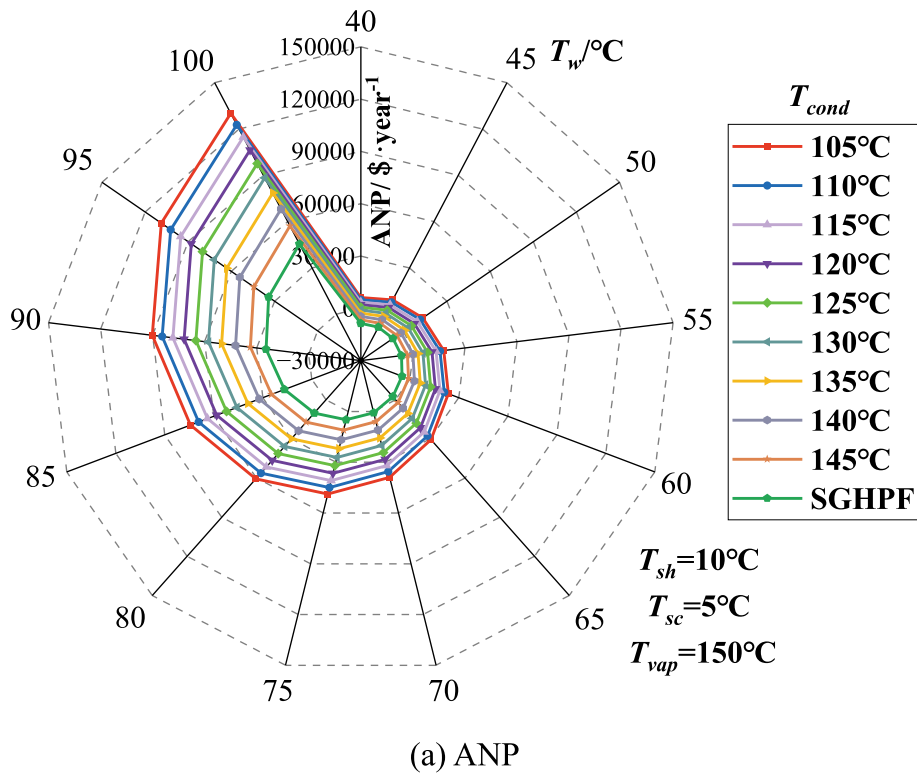


Fig. 14 Comparison of ANP and PBP of SGHPF and SGHPO with different condensation temperature

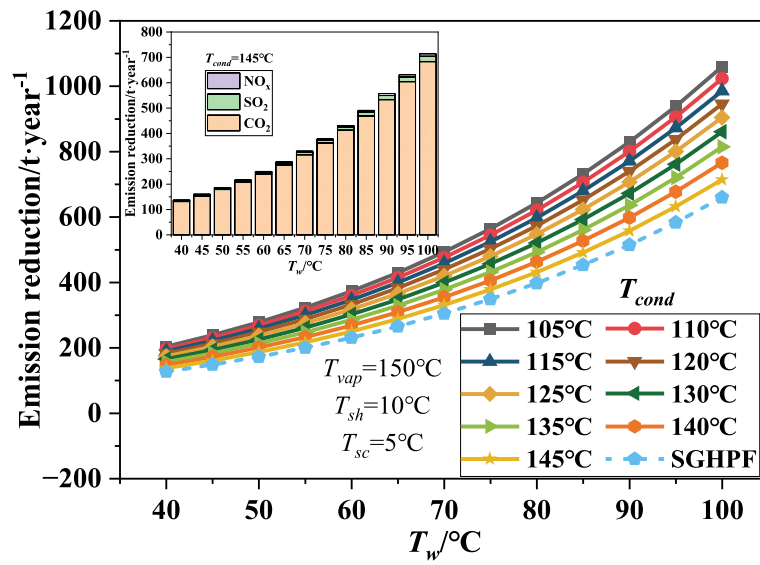


Fig. 15 Comparison of annual emission reduction of SGHPF and SGHPO with different condensation temperature

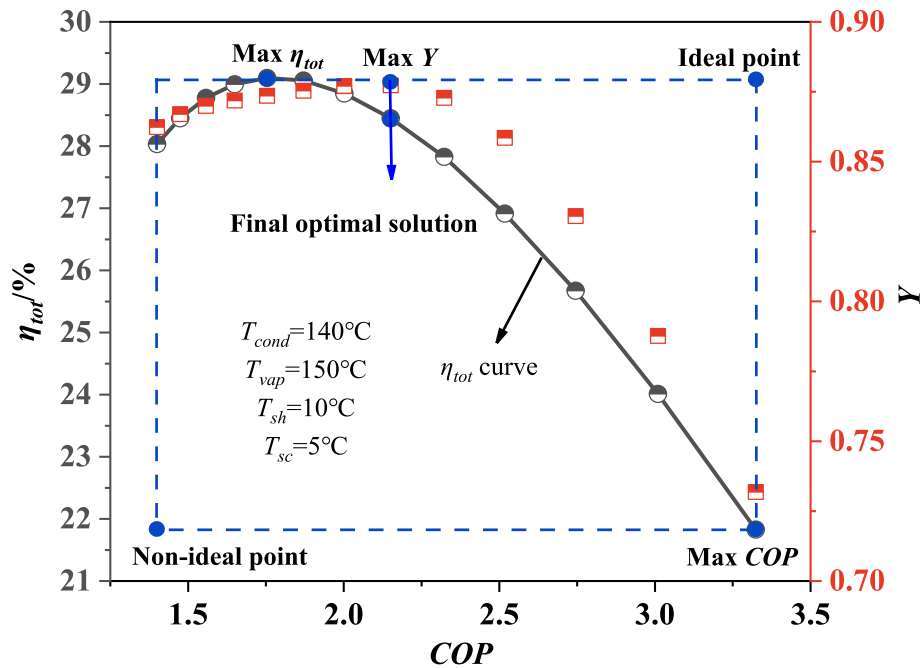


Fig. 16 Evaluation indicator balance results for SGHPO with condensation temperature of 140 °C

- The 4E performance of SGHPO with different condensation temperatures is higher than that of SGHPF when the same temperature steam is generated.
- SGHPO increased COP, exergy efficiency, ANP and emission reductions of pollutants as the con-

densation temperature decreased, which was more favorable to the system performance. At the most unfavorable operating condition of the system, the COP minimum value is 1.31 and the exergy efficiency minimum value is 20.42%. The lowest condensation temperature that produces a PBP is 125

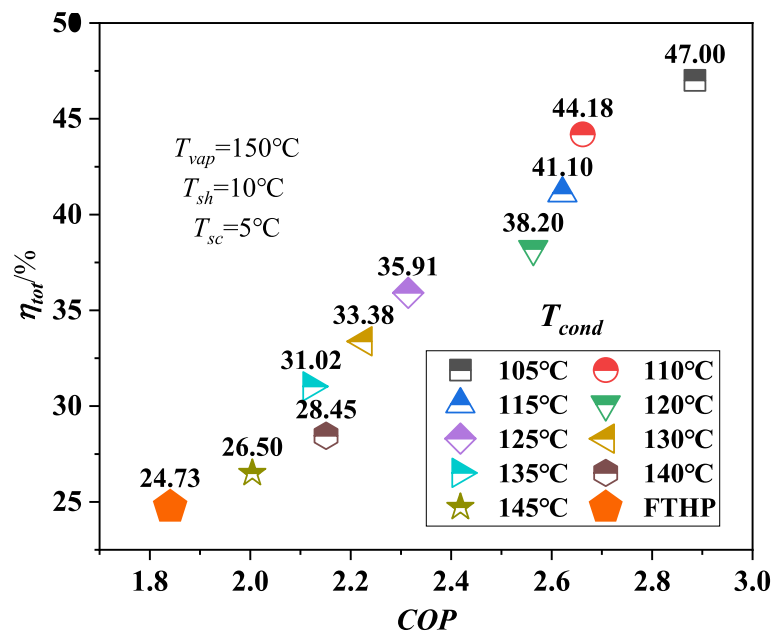


Fig. 17 Evaluation indicator balance results for SGHPO and SGHPF

		Waste heat temperature/°C												
		40	45	50	55	60	65	70	75	80	85	90	95	100
Condensation temperature/°C	105	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	110	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	115	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	120	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	125	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	130	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	135	N	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	140	N	N	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	145	N	N	N	N	N	Y	Y	Y	Y	Y	Y	Y	Y
SG HPF		N	N	N	N	N	N	Y	Y	Y	Y	Y	Y	Y

Fig. 18 Recommended operating conditions for SGHP

°C at a heat source temperature of 40 °C, at which case the PBP is 1.33 years.

- As the heat source temperature varies, an optimal balance value between COP and exergy efficiency

of SGHP emerges when the generated steam temperature remains constant. The range of heat source temperature required to reach the optimum values of SGHPO with different condensation tempera-

tures and SGHPF is identified as 70°C to 80°C. The optimized maximum COP and exergy efficiency are found to be 2.89 and 47.00%, respectively

6 Nomenclature

ANP Annual net profit

COP Coefficient of performance

CI Capital initial investment

CRF Capital recovery factor

E Exergy

EAC Equivalent annual cost

GWP Global warming potential

h Enthalpy

HP Heat pump

I Exergy destruction

IHX Intermediate heat exchanger

M Mass flow rate

N Operation times of a year

NBP Normal boiling point

ODP Ozone depletion potential

OM Operation and Maintenance

P Pressure

PBP Payback period

Q Heating

s Entropy

SG Safety group classification

SGHP Steam generating heat pump

T Temperature

W Power consumption

Y Proximity index

6.1 Greek symbols

Π Compressor pressure ratio

η Efficiency

φ Maintenance factor

ψ Compressor isentropic efficiency

6.2 Subscripts

0 Environmental status point

cond Condenser

evap Evaporator

f Refrigerant

gas CO₂, SO₂ and NO_x

in Inlet

k Each component

out Outlet

per Per unit of steam

refc Refrigerant compressor

sc Subcooling degree

sh Suction superheat

tot Total

tv Throttling valve

vap Steam

vapc Water vapor compressor

w Waste heat

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Authors' contributions

Xudong Ma: Software, Validation, Formal analysis, Writing-Original Draft. Yuting Wu: Methodology, Investigation, Resources, Writing-Review. Yanjun Du: Conceptualization, Writing-Review & Editing, Funding acquisition. Cancan Zhang: Writing-Review & Editing, Visualization. Biao Lei: Writing-Review. Yuanwei Lu: Writing-Review.

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Data availability

The original data are available from corresponding authors upon reasonable request.

Declarations

Ethics approval and consent to participate

The manuscript is original and not submitted to other journal, all the results are without fabrication. The authors consent to participate in the manuscript.

Consent for publication

All participants have provided consent for the research to be submitted for publication.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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